University of Warwick
MSc Business Analytics

IB9LQ0 Generative AI and AI Applications
2025-2026

Individual Assignment 2

Retrieval-Augmented Generation for Sustainability Reporting Standards

Table of Contents

1. Introduction

Large language models answer fluently but from fixed, dated training data, making them unreliable for regulatory questions that depend on exact wording and on rules that change over time. Retrieval-augmented generation (RAG) addresses this by retrieving relevant source text at query time and grounding the answer in it.

This report presents a RAG system over sustainability-reporting standards and evaluates whether it outperforms a plain language model. Three pipelines are compared on identical questions:

- **No-RAG baseline** - the language model answers directly, with no retrieval.
- **Baseline RAG** - standard dense retrieval over fixed-size chunks with a plain prompt.
- **Enhanced RAG** - several recent retrieval techniques combined with a grounded, citation-enforcing prompt.

The system is assessed on both retrieval and answer quality across a diverse query set, including questions answerable only from a regulation published after the model's training cut-off. The report sets out the design decisions, results, evaluation, and a reflection on failure modes and limitations.

2. Domain and motivation

The domain is European Sustainability Reporting Standards (ESRS) and the global ISSB standards (IFRS S1 and S2). Compliance reporting is an unusually strong fit for RAG, for three reasons:

- **Recency** - the corpus includes Delegated Regulation (EU) 2025/1416, the November 2025 “Quick Fix” amending the ESRS phase-in timetable, which postdates every current model's training cut-off, so a plain model cannot know it.
- **Precision and traceability** - compliance answers turn on the exact wording of dense, acronym-heavy provisions and must cite their source; a model paraphrasing from memory cannot reliably ground or attribute its claims, and an invented answer carries real regulatory risk.
- **Vocabulary gap** - practitioners ask plain-English questions (“do smaller firms get more time?”) while the standards use defined terms (“undertakings”, “phasing-in”, “Article 5(2)”), so naive keyword matching fails.

The knowledge base is five official PDFs: IFRS S1, IFRS S2, ESRS 1 General Requirements, the consolidated ESRS Set 1 annex, and the 2025 Quick Fix. This mix is deliberate: the 2023 standards are familiar to the model while the 2025 amendment is genuinely unseen, letting the evaluation separate what RAG recalls from what it supplies anew.

3. System architecture

Three pipelines are compared on identical questions. The no-RAG baseline calls the LLM directly with no context. The baseline RAG arm uses fixed-size character chunking (1000 characters, 150 overlap), dense retrieval over BAAI/bge-small-en-v1.5 embeddings in ChromaDB, and a plain prompt. The enhanced RAG arm adds four modern techniques:

- **Structure-aware chunking** - splits on the standards' own paragraph numbering and headings, keeping each provision coherent (2,639 chunks versus the baseline's 1,455).
- **Query rewriting / HyDE** - reformulates the plain-English question into standards terminology before searching, targeting the vocabulary gap.
- **Hybrid retrieval with Reciprocal Rank Fusion** - runs BM25 keyword search and dense semantic search in parallel and fuses their rankings, so exact-term matches (paragraph identifiers, defined terms) and paraphrased semantic matches reinforce one another.
- **Cross-encoder reranking** (ms-marco-MiniLM-L-6-v2) - jointly scores each query-chunk pair to sharpen the final top-k, correcting cases where fusion ranks a loosely related chunk too highly.

Generation uses a grounded prompt requiring a bracketed citation for every claim and an explicit refusal when the context does not support an answer.

3.1 Model choice

The LLM throughout is Llama-3.1-8b-instant, served via the Groq API. This choice was deliberate: an open-weight model keeps the pipeline reproducible and free of per-token cost, and Groq's inference runs the full evaluation in minutes. The 8B size was chosen over a 70B alternative for a practical reason found during development, the larger model exhausted the free-tier daily token allowance mid-run, while the 8B model's higher request limit let the suite complete reliably. A compact model is also a more honest baseline for the post-training argument: if retrieval alone lifts it from 1.0/5 to 5.0/5 on unseen regulation, RAG's value is shown without leaning on a large model's latent knowledge. The caveat that the same 8B model also acts as judge is examined in the reflection. Embeddings and reranking run locally, so the LLM is the only external call.

4. Evaluation method

The test set is 20 queries spanning the types named in the brief: simple-fact, deep-context, ambiguous plain-English, edge/refusal, and post-training. Six target the 2025 Quick Fix (answerable only from that document); two are out-of-corpus refusal probes that should be declined rather than answered. Each query carries gold passages (distinctive substrings identifying a relevant chunk) and a reference answer. A single relevance rule drives both retrieval metrics: a chunk is relevant if it comes from the gold source and contains a gold passage. Retrieval is scored by precision@k, true recall@k (relevant chunks retrieved divided by the total relevant chunks in the store, with the denominator precomputed per collection so recall is a true proportion rather than a hit-rate proxy), and mean reciprocal rank (MRR). Generation is scored

1-5 on correctness, grounding and completeness by a zero-temperature LLM judge. A coverage diagnostic runs before scoring to confirm every in-corpus query has at least one relevant chunk in each store, guarding against a broken recall denominator.

5. Results

RAG beats the plain LLM decisively, and the post-training queries are the clearest proof. On the six Quick Fix queries, no-RAG averaged just 1.0/5 correctness while both RAG arms scored 5.0/5, and grounding rose from 3.0 to 5.0. Asked about the Quick Fix relief for anticipated-financial-effects disclosures, no-RAG replied it was "not aware of any information about the 2025 Quick Fix amendments", whereas the enhanced arm retrieved the regulation and stated the precise relief (omit ESRS 2 SBM-3 48(e) for the first year, three years for Article 5(2) undertakings). Across all queries, correctness climbed from 3.60 to 4.65 to 4.80; grounding from 4.1 to 4.8 to 4.9; completeness from 4.0 to 4.6 to 4.8 (Table 1, Appendix). The ordering is consistent across all three metrics; the post-training contrast is in Table 2.

On retrieval, the enhanced arm improved recall@k (0.134 to 0.188) and MRR (0.460 to 0.535), but precision@k was slightly lower (0.36 to 0.30) - an expected trade-off, not a defect. Hybrid retrieval with RRF casts a wider net, surfacing relevant provisions the baseline misses and ranking the first relevant hit higher (lifting recall and MRR), but it admits more borderline chunks into the top-k, diluting precision. The structure-aware store also changes the recall denominator, as paragraph-level splitting creates more, smaller chunks. The enhanced retriever is therefore better at what matters for compliance, finding the right provision and ranking it first rather than uniformly superior on every metric.

The per-type breakdown (Table 3, Appendix) supports this: enhanced retrieval is strongest on deep-context (precision 0.65) and post-training (recall 0.49) queries, where precise terminology and keyword matching pay off, and weakest on ambiguous queries (recall 0.01), where novice phrasing is hardest to match to regulatory language. The techniques help most where the domain is hardest for a plain model, and least where the question is under-specified.

6. Failure modes and reflection

Four observations stand out:

● **Grounding can over-fire**- on the deep-context query comparing ESRS and ISSB materiality, the enhanced arm refused ("the provided standards text does not cover this") although the topic is well covered, because hybrid retrieval did not surface a chunk it judged sufficient and the strict prompt then forbade an answer. This false refusal trades a few correct answers for a near-guarantee against fabrication often defensible in compliance, since a confident wrong citation is worse than an honest "not found", but it would be tuned (relaxing the refusal threshold or widening k) before deployment.

● **Retrieval failure caps generation**- where precision@k was zero, no prompt sophistication recovered the answer, confirming generation quality is bounded by retrieval quality.

● **The base model hallucinates confidently**- swapping the names of IFRS S1 and S2 and inventing the "four content areas", authoritative-looking errors that would mislead a non-expert; RAG corrected both.

● **Refusal on out-of-corpus probes worked**, confirming the prompt distinguishes "absent from these documents" from "unknown to the model".

6.1 Limitations

The main methodological limitation is judge saturation. Generation is scored by the same Llama-3.1-8b model that generates answers, and it is lenient, most arms cluster near 5/5, so the generation metrics cannot finely separate baseline from enhanced and should be read as a ceiling check, not a precise ranking. The retrieval metrics are the more trustworthy signal; a stronger independent or human judge would sharpen the comparison. Practical limitations include an in-memory vector store rebuilt each session (not production-scalable), added latency from the hybrid-plus-rerank pipeline, and a fixed corpus snapshot needing re-ingestion as regulation evolves - the very property that motivated the project.

7. Conclusion

The system closes a genuine knowledge gap the plain LLM cannot, proven on regulation published after the model's cut-off, and the enhanced techniques deliver measurable retrieval gains in recall and ranking for a modest precision trade-off, while grounding enforces honest citations and refusals. The practical lesson is that for compliance RAG, retrieval quality and disciplined grounding matter more than generation fluency, and evaluation must look past saturated generation scores to the retrieval metrics and failure cases that distinguish a usable system from a plausible one.

Appendix

Table 1. Mean scores across all 20 queries.

Arm	Prec@k	Recall@k	MRR	Correct.	Ground./Compl.
No-RAG	—	—	—	3.60	4.1 / 4.0
Baseline RAG	0.36	0.134	0.460	4.65	4.8 / 4.6
Enhanced RAG	0.30	0.188	0.535	4.80	4.9 / 4.8

Table 2. Post-training (2025 Quick Fix) queries: base LLM vs RAG.

Arm (Quick Fix queries only)	Correctness	Groundedness
No-RAG	**1.0**	3.0
Baseline RAG	**5.0**	5.0
Enhanced RAG	**5.0**	5.0

Table 3. Per-type retrieval and generation scores (enhanced arm).

Query type	Prec@k	Recall@k	Correct.	Ground.
simple_fact	0.25	0.28	4.0	4.5
deep_context	0.65	0.18	5.0	5.0
ambiguous	0.20	0.01	5.0	5.0
post_training	0.33	0.49	5.0	5.0
edge_not_in_corpus	0.00	—	5.0	5.0
edge_out_of_scope	0.00	—	5.0	5.0